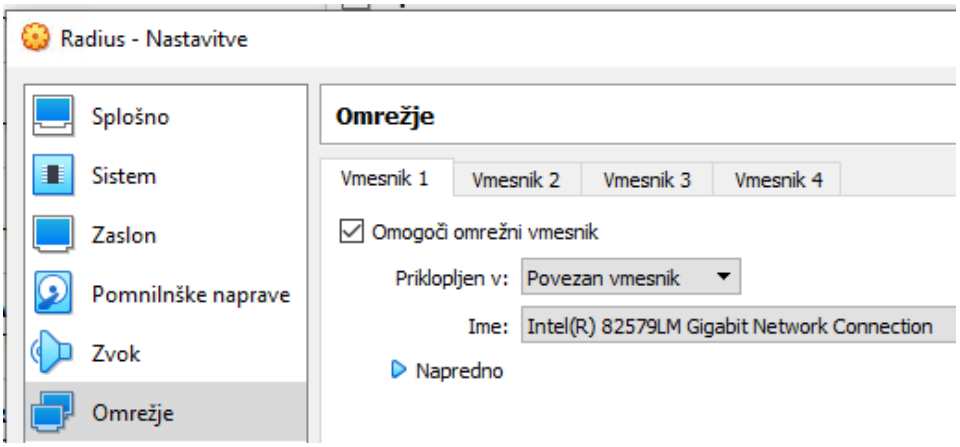
V navodilih je opisana namestitev strežnika Radius, njegove odprtokodne različice FreeRadius ter orodja DaloRadius, ki je grafični vmesnik za nadzor/vzdrževanje delovanja strežnika Radius.

- **Potreben je "čisti" CentOS VBox gost – samo osnovna namestitev, brez nameščenih storitev.**
- **Strežniku CentOS dodelite za EDINO mrežno kartico "Povezan vmesnik".**
- **Strežniku dodelite 8 GB delovnega pomnilnika.**



Za delovanje strežnika Radisu potrebujemo:

- Freeradius
- Podatkovno bazo MariaDb.
- Skriptni jezik PHP 7 in spletni strežnik Apache (httpd).
- Orodje DaloRadius.

Potrebno bo konfigurirati baze, mape, pravice dostopa, konfiguracijske datoteke strežnika Radius, nastaviti požarni zid in omogočiti dostop do Radius strežnika preko SeLinuxa.

KAZALO VSEBINE NAVODIL:

1 Freeradius 2
2 MariaDB 3
3 PHP in httpd 6
4 Konfiguracija FreeRadius-a 7
5 Firewall 8
6 DaloRadius 11
7 Nastavitve usmerjevalnika 16

1 Freeradius

Najprej je potrebno posodobiti sistem.

yum update -y

Namestitev potrebnih paketov za FreeRadisu, MaridoDB in httpd:

sudo yum search all freeradius

sudo yum -y install freeradius freeradius-utils freeradius-mysql freeradius-perl

Omogočiti in startati je potrebno samo storitev freeradius:

systemctl start radiusd.service

systemctl enable radiusd.service

Preverjanje delovanja:

systemctl status radiusd.service

```
[root@localhost ~]# systemctl status radiusd.service
● radiusd.service - FreeRADIUS high performance RADIUS server.
   Loaded: loaded (/usr/lib/systemd/system/radiusd.service; enabled; vendor pre>
   Active: active (running) since Tue 2020-03-03 22:43:56 CET; 45s ago
 Main PID: 9408 (radiusd)
    Tasks: 6 (limit: 26213)
   Memory: 9.3M
   CGroup: /system.slice/radiusd.service
           └─9408 /usr/sbin/radiusd -d /etc/raddb

mar 03 22:43:56 localhost.localdomain systemd[1]: Starting FreeRADIUS high perf>
mar 03 22:43:56 localhost.localdomain systemd[1]: Started FreeRADIUS high perfo>
lines 1-11/11 (END)
```

2 MariaDB

Izbira različice:

gedit /etc/yum.repos.d/MariaDB.repo

[mariadb]
name = MariaDB
baseurl = http://yum.mariadb.org/10.5.1/centos7-amd64
gpgkey=https://yum.mariadb.org/RPM-GPG-KEY-MariaDB
gpgcheck=1

Shranite zgornje nastavitve!

yum -y update
(za uveljavitev novega "repositorija")

yum install -y mariadb-server mariadb

systemctl start mariadb

systemctl enable mariadb

systemctl status mariadb

```
[root@localhost ~]# systemctl status mariadb
● mariadb.service - MariaDB 10.3 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; vendor pre>
   Active: active (running) since Tue 2020-03-03 22:47:10 CET; 9s ago
     Docs: man:mysqld(8)
           https://mariadb.com/kb/en/library/systemd/
 Main PID: 10370 (mysqld)
   Status: "Taking your SQL requests now..."
    Tasks: 30 (limit: 26213)
   Memory: 81.7M
   CGroup: /system.slice/mariadb.service
           └─10370 /usr/libexec/mysqld --basedir=/usr

mar 03 22:47:09 localhost.localdomain mysql-prepare-db-dir[10267]: Please repor>
mar 03 22:47:09 localhost.localdomain mysql-prepare-db-dir[10267]: The latest i>
mar 03 22:47:09 localhost.localdomain mysql-prepare-db-dir[10267]: You can find>
mar 03 22:47:09 localhost.localdomain mysql-prepare-db-dir[10267]: http://dev.m>
mar 03 22:47:09 localhost.localdomain mysql-prepare-db-dir[10267]: Consider joi>
mar 03 22:47:09 localhost.localdomain mysql-prepare-db-dir[10267]: https://mari>
mar 03 22:47:09 localhost.localdomain mysqld[10370]: 2020-03-03 22:47:09 0 [Not>
mar 03 22:47:09 localhost.localdomain mysqld[10370]: 2020-03-03 22:47:09 0 [War>
mar 03 22:47:09 localhost.localdomain mysqld[10370]: 2020-03-03 22:47:09 0 [War>
```

Prva konfiguracija mariadb (na začetku ni gesla!):

mysql_secure_installation

```
[root@localhost ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user.  If you've just installed MariaDB, and
you haven't set the root password yet, the password will be blank,
so you should just press enter here.

Enter current password for root (enter for none):
```

Nastavite svoje geslo (na začetku ga ni, potrdite z Enter, da ni gesla)

Nato nastavite novo geslo, odstranite možnost dostopa za anonimne uporabnike, preprečite oddaljeni dostop, odstranite testno bazo in ponovno naložite tabele s privilegiji.

Set the root password (vpišite ***centos*** za novo geslo)
Remove the Anonymous User
Disable root remote login
Remove Test Database
Reload Privilege Tables now

```
Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password ensures that nobody can log into the MariaDB
root user without the proper authorisation.

Set root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
 ... Success!


By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them.  This is intended only for testing, and to make the installation
go a bit smoother.  You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
 ... Success!
```

```
Normally, root should only be allowed to connect from 'localhost'.  This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
 ... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access.  This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
 - Dropping test database...
 ... Success!
 - Removing privileges on test database...
 ... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] y
 ... Success!

Cleaning up...
```

Dovolite dostop do baze samo lokalno:

gedit /etc/my.cnf.d/mariadb-server.cnf

Popravite v sekciji [mysqld]:

[mysqld]
bind-address=127.0.0.1

```
Dejavnosti   Gedit - urejevalnik besedil   mar 3 22:55
Odpri                *mariadb-server.cnf
                     /etc/my.cnf.d
# customize your systemd unit file for mysqld/mariadb according to the
# instructions in http://fedoraproject.org/wiki/Systemd
[mysqld]
datadir=/var/lib/mysql
socket=/var/lib/mysql/mysql.sock
log-error=/var/log/mariadb/mariadb.log
pid-file=/run/mariadb/mariadb.pid
bind-address=127.0.0.1

#
# * Galera-related settings
#
[galera]
```

Ponoven zagon mariadb in radiusa:

systemctl restart mariadb

systemctl restart radiusd.service

3 PHP in httpd

Namestitev httpd:

dnf -y install httpd httpd-devel

systemctl enable httpd

systemctl start httpd

Namestitev PHP – različice 7.3:

cd ~

dnf install https://dl.fedoraproject.org/pub/epel/epel-release-latest-8.noarch.rpm

Preveri:
rpm -qa | grep epel

```
[root@localhost ~]# rpm -qa | grep epel
epel-release-8-7.el8.noarch
[root@localhost ~]#
```

dnf install https://rpms.remirepo.net/enterprise/remi-release-8.rpm

Preverjanje:

rpm -qa | grep remi-release

```
[root@localhost ~]# rpm -qa | grep remi-release
remi-release-8.0-4.el8.remi.noarch
```

Namestitev PHP-ja (na voljo so različice 7.2, 7.3 in 7.4 – izbrali bomo 7.3):

dnf module enable php:remi-7.3

dnf install php php-cli php-common

Še potrebne knjižnice:

dnf -y install php-{cli,curl,mysqlnd,gd,pear,mbstring,xml,pear}

Preverjanje različice PHP-ja:

php -v

Ponoven zagon httpd (zaradi php-ja):

systemctl restart httpd

4 Konfiguracija FreeRadius-a

Nastavitev oz. kreiranje baze, uporabnika, gesla in pravic (baza *radius*, uporabnik *radius*, geslo *radius123+*):

mysql -u root -p

CREATE DATABASE radius;
GRANT ALL ON radius.* TO radius@localhost IDENTIFIED BY "radius123+";
FLUSH PRIVILEGES;
exit;

Spreminjanje konfiguracije:

mysql -uroot -pcentos radius < /etc/raddb/mods-config/sql/main/mysql/schema.sql

ln -s /etc/raddb/mods-available/sql /etc/raddb/mods-enabled/

Spremenite naslednje možnosti (razlaga v nadaljevanju pod prikazano skripto):

gedit /etc/raddb/mods-available/sql

Spremeni:

1. Spremenite **driver = "rlm_sql_null"** v **driver = "rlm_sql_mysql"**
2. Spremenite **dialect = "sqlite"** v **dialect = "mysql"**
3. Odstranite komentar (# na začetku vrstice)) pri **server**, **port**, **login** in **password** in spremenite:

   server = "localhost"
   port = 3306
   login = "radius"
   password = "radius123+"

   Ime baze je že nastavljeno na ime "radius" (preverite):

   radius_db = "radius"

4. Odstranite komentar (#) pri **read_clients = yes**
5. **Shranite nastavitve.**

Zamenjati je potrebno skupino (*group*) za mapo SQL:

chgrp -h radiusd /etc/raddb/mods-enabled/sql

5 Firewall

Požarni zid – potrebno je dovoliti dostop preko njega:

systemctl start firewalld

firewall-cmd --get-services | egrep 'http|https|radius'

firewall-cmd --add-service={http,https,radius} --permanent

firewall-cmd --reload

Test delovanja:

firewall-cmd --get-default-zone

firewall-cmd --list-services --zone=public

```
[root@localhost ~]# firewall-cmd --list-services --zone=public
cockpit dhcpv6-client http https radius ssh
[root@localhost ~]#
```

Če je radius deluje v redu, se mora pognati v "debug-mode" načinu:

pkill radiusd

Debug mode za zagon radius-a:

radiusd -X

Prikaz bi moral biti naslednji (brez napak):

```
Listening on auth address * port 1812 bound to server default
Listening on acct address * port 1813 bound to server default
Listening on auth address :: port 1812 bound to server default
Listening on acct address :: port 1813 bound to server default
Listening on auth address 127.0.0.1 port 18120 bound to server inner-tunnel
Listening on proxy address * port 58382
Listening on proxy address :: port 47273
Ready to process requests
```

Test se konča s CTRL+C in storitev se mora zagnati v "normalnem" načinu:

pkill radiusd

systemctl restart radiusd

Kreiranje uporabnikov (clients)

Popraviti je potrebno nastavitve v datoteki clients.conf:

gedit /etc/raddb/clients.conf

V vrstici 42 popravite:

Ipaddr = *

```
#
#  If both A and AAAA records are found, A records will be
#  used in preference to AAAA.
ipaddr = *

#  Same as ipaddr but allows v4 addresses only. Requires A
#  record for domain names.
```

V vrstici 100 preverite nastavitve za opcijo *secret* – v vaji pustite testing123 (za "resno uporabo" obvezno spremenite besedo):

```
#  The default secret
#  not be used in any
#
secret = testing123
```

Zapustite urejevalnik (gedit) in shranite spremembe v datoteki clients.conf.

Kreiranje certifikata

V datoteki ca.conf:

gedit /etc/raddb/certs/ca.cnf

Popravite ter shranite naslednji odsek (na sliki) – v realnosti vpišite veljavne podatke:

```
[certificate_authority]
countryName             = SL
stateOrProvinceName     = Radius
localityName            = Somewhere
organizationName        = Example Inc.
emailAddress            = admin@example.com
commonName              = "Example Certificate Authority"
```

V datoteki client.cnf:

gedit /etc/raddb/certs/client.cnf

Popravite ter shranite naslednji odsek (na sliki) – v realnosti vpišite veljavne podatke:

```
[client]
countryName             = SL
stateOrProvinceName     = Radius
localityName            = Somewhere
organizationName        = Example Inc.
emailAddress            = user@example.com
commonName              = user@example.com
```

V datoteki server.cnf:

gedit /etc/raddb/certs/server.cnf

Popravite sekcijo na sliki – v realnosti vpišite veljavne podatke:

```
[server]
countryName             = SL
stateOrProvinceName     = Radius
localityName            = Somewhere
organizationName        = Example Inc.
emailAddress            = admin@example.com
commonName              = "Example Server Certificate"
```

Zaženite naslednje ukaze za kreiranje certifikata:

cd /etc/raddb/certs

rm -f *csr *key

make

6 DaloRadius

Namestiti je potrebno DaloRadius:

cd ~

wget https://github.com/lirantal/daloradius/archive/master.zip

unzip master.zip

mv daloradius-master/ daloradius

cd daloradius

Uvoziti je potrebno MySQL bazo oz. tabele (pri zahtevi za geslo vpišite *centos*):

mysql -u root -p radius < contrib/db/fr2-mysql-daloradius-and-freeradius.sql

mysql -u root -p radius < contrib/db/mysql-daloradius.sql

Popraviti je potrebno še pravice:

cd ..

mv daloradius /var/www/html/

chown -R apache:apache /var/www/html/daloradius/

chmod 664 /var/www/html/daloradius/library/daloradius.conf.php

Popraviti je potrebno konfiguracijsko datoteko programa DaloRadius:

gedit /var/www/html/daloradius/library/daloradius.conf.php

Popravite pri naslednjih treh nastavitvah podatke (kot ste jih vnesli)

CONFIG_DB_USER
CONFIG_DB_PASS
CONFIG_DB_NAME

```
$configValues['DALORADIUS_VERSION'] = '1.1-3 BETA';
$configValues['DALORADIUS_DATE'] = '06 Dec 2019';
$configValues['FREERADIUS_VERSION'] = '2';
$configValues['CONFIG_DB_ENGINE'] = 'mysqli';
$configValues['CONFIG_DB_HOST'] = 'localhost';
$configValues['CONFIG_DB_PORT'] = '3306';
$configValues['CONFIG_DB_USER'] = 'radius';
$configValues['CONFIG_DB_PASS'] = 'radius123+';
$configValues['CONFIG_DB_NAME'] = 'radius';
$configValues['CONFIG_DB_TBL_RADCHECK'] = 'radcheck';
```

Dovoljenje za SeLinux (drugače ga je potrebno ustaviti):

semanage fcontext -a -t httpd_sys_rw_content_t "/var/www/html/daloradius(/.*)?"

Pri vsaki spremembi nastavitev oz. datotek (npr. pri posodobitvah) je <u>potrebno vedno znova izvesti ta ukaz</u>, s katerim se dovoli dostop preko SeLinuxa!

restorecon -Rv /var/www/html/daloradius/*

Še zadnja ukaza za zaključek namestitve Dalraiusa:

sudo pear install DB

sudo pear install MDB2

Ponoven zagon potrebnih storitev:

systemctl restart radiusd

systemctl restart mariadb

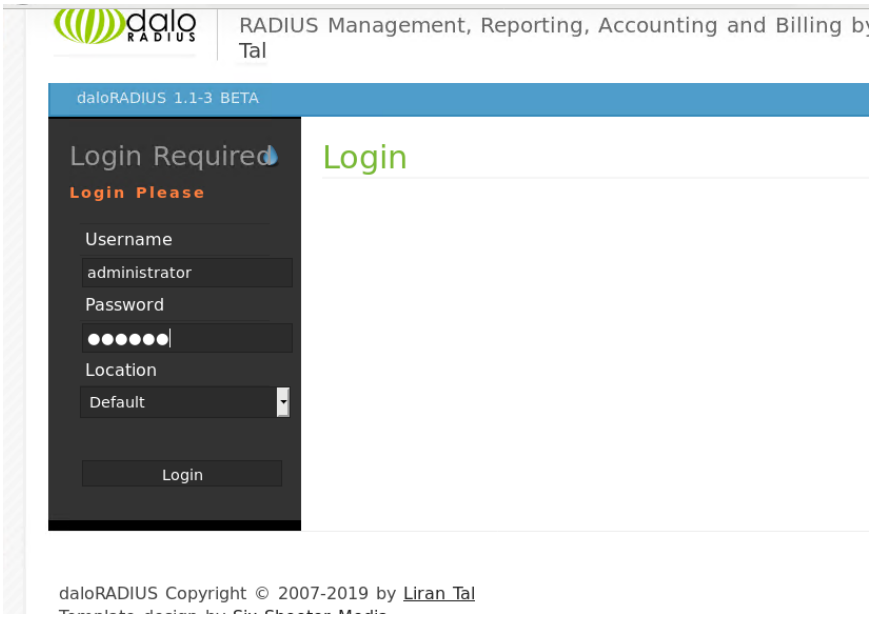
systemctl restart httpd

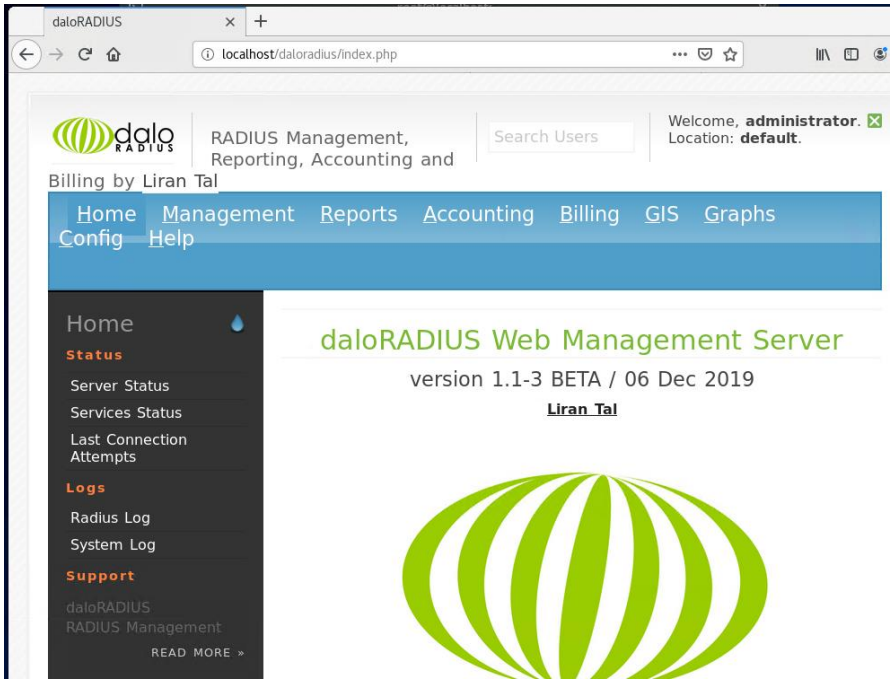
Preizkus delovanja:

http://IP_naslov_strežnikaCentOS/daloradius/login.php

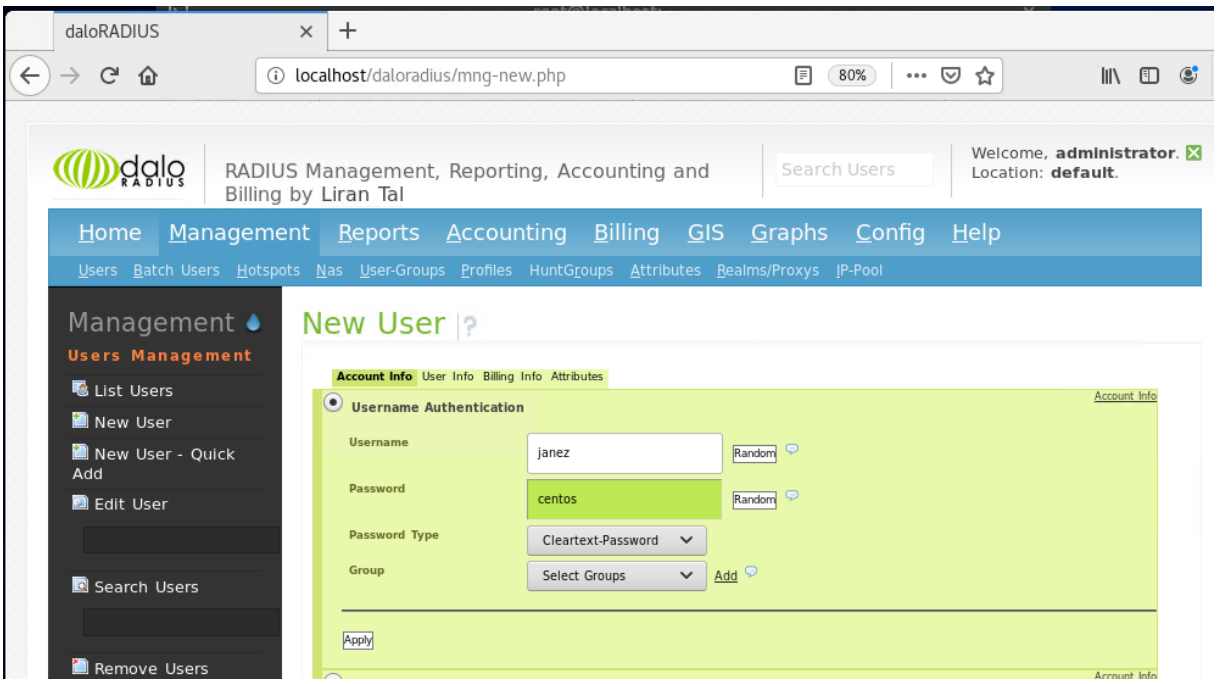
Privzeti podatki za uporabnika in geslo:

Username: **administrator**
Password: **radius**

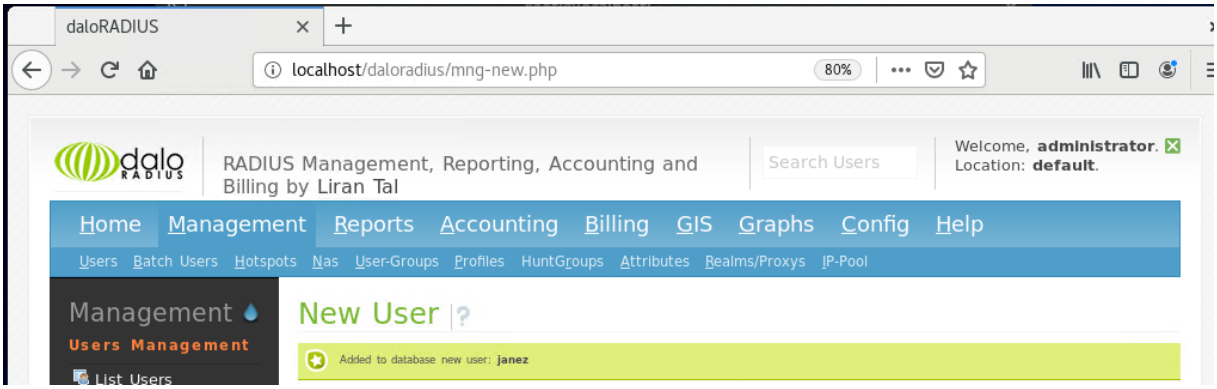




Dodajte novega uporabnika (*Management/New User*, uporabnik *janez*, geslo naj bo *centos*):

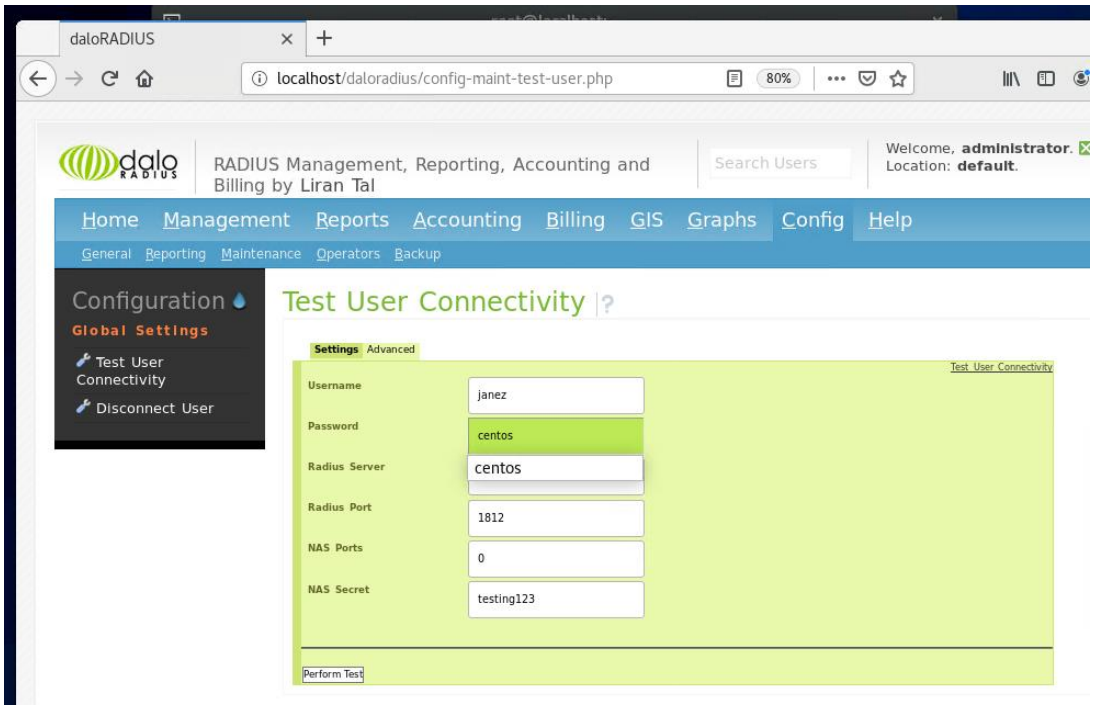


Apply za potrditev

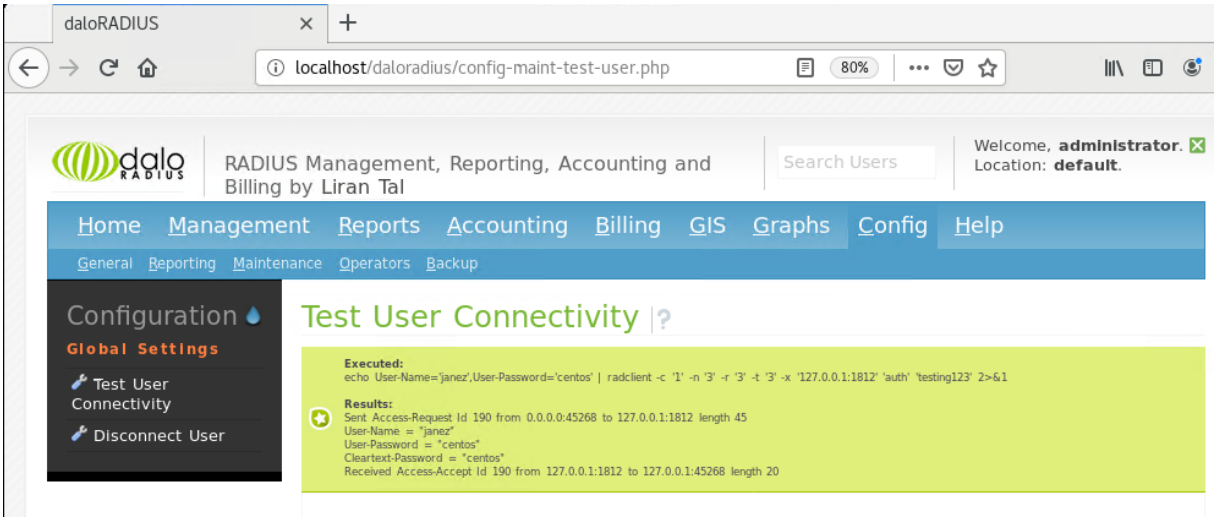


Testiranje (Config/Maintenance/Test user connectivity)

Vpišite uporabnika in izberite Perform Test:



Odziv testiranja – deluje, če je odziv kot je na sliki spodaj (*Received Access-Accept id …*)



7 Nastavitve usmerjevalnika

Nastaviti je potrebno v Wireless Security Settings:

WPA/WPA2 – **Enterprise**

Authentication Type: **Auto**
Encryption: **Auto**
Radius Server IP**:** (**VPIŠITE NASLOV IP STREŽNIKA RADIUS**)
Radius Server Port: **1812**
Radius Server Password: **testing123**

Pri prijavi (v Firefox-u) morate potrditi certifikat (samo prvič) !